PGCP-AC AUGUST 26
Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:
   - Books -> book_id,title,author_id,published_year,available_copies
   - Authors -> author_id,name,country
   - Members -> member_id,name,email,phone
   - Borrow_Records -> record_id,member_id,book_id,borrow_date,return_date

2. Write the SQL command to create a table Authors with the following fields:
   - author_id (Primary Key, INT)
   - name (VARCHAR(100))
   - country (VARCHAR(50))

Answer :

```
mysql> CREATE TABLE AUTHORS (AUTHOR_ID INT PRIMARY KEY,NAME VARCHAR(100),COUNTRY VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)

mysql> DESC AUTHORS;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| AUTHOR_ID | int          | NO   | PRI | NULL    |       |
| NAME      | varchar(100) | YES  |     | NULL    |       |
| COUNTRY   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

3. Write the SQL command to create a table Books with the following fields:
   - book_id (Primary Key, INT)
   - title (VARCHAR(150))
   - author_id (Foreign Key referencing Authors)
   - published_year (YEAR)
   - available_copies (INT)

Answer:

```
mysql> CREATE TABLE BOOKS
    -> (BOOK_ID INT PRIMARY KEY,TITLE VARCHAR(150),
    -> AUTHOR_ID INT ,CONSTRAINT FK_AUTHORID FOREIGN KEY (AUTHOR_ID) REFERENCES AUTHORS(AUTHOR_ID),
    -> PUBLISHED_YEAR YEAR,AVAILABLE_COPIES INT);
Query OK, 0 rows affected (0.09 sec)

mysql> DESC BOOKS;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| BOOK_ID          | int          | NO   | PRI | NULL    |       |
| TITLE            | varchar(150) | YES  |     | NULL    |       |
| AUTHOR_ID        | int          | YES  | MUL | NULL    |       |
| PUBLISHED_YEAR   | year         | YES  |     | NULL    |       |
| AVAILABLE_COPIES | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

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4. Write the SQL command to create a table Members with:
- member_id (Primary Key, INT)
- name (VARCHAR(100))
- email (VARCHAR(100), unique)
- phone (VARCHAR(15))

Answer:

```
mysql> CREATE TABLE MEMBERS (MEMBER_ID INT PRIMARY KEY,NAME VARCHAR(100),EMAIL VARCHAR(100)
UNIQUE,PHONE VARCHAR(15));
Query OK, 0 rows affected (0.08 sec)

mysql> DESC MEMBERS;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| MEMBER_ID | int          | NO   | PRI | NULL    |       |
| NAME      | varchar(100) | YES  |     | NULL    |       |
| EMAIL     | varchar(100) | YES  | UNI | NULL    |       |
| PHONE     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

5. Write the SQL command to create a table Borrow_Records with:
- record_id (Primary Key, INT)
- member_id (Foreign Key referencing Members)
- book_id (Foreign Key referencing Books)
- borrow_date (DATE)
- return_date (DATE)

Answer:

```
mysql> CREATE TABLE BORROW_RECORDS (RECORD_ID INT PRIMARY KEY,
    -> MEMBER_ID INT,CONSTRAINT FK_MEMBERID FOREIGN KEY (MEMBER_ID) REFERENCES MEMBERS(MEMBER_ID),
    -> BOOK_ID INT,CONSTRAINT FK_BOOKID FOREIGN KEY (BOOK_ID) REFERENCES BOOKS(BOOK_ID),
    -> BORROW_DATE DATE,RETURN_DATE DATE);
Query OK, 0 rows affected (0.09 sec)

mysql> DESC BORROW_RECORDS;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| RECORD_ID   | int  | NO   | PRI | NULL    |       |
| MEMBER_ID   | int  | YES  | MUL | NULL    |       |
| BOOK_ID     | int  | YES  | MUL | NULL    |       |
| BORROW_DATE | date | YES  |     | NULL    |       |
| RETURN_DATE | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

6. Modify the Books table to add a column genre of type VARCHAR(50).

Answer:

```
mysql> ALTER TABLE BOOKS ADD COLUMN GENRE VARCHAR(50);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

7. Write the SQL command to drop the Borrow_Records table.

Answer:

```
mysql> DROP TABLE BORROW_RECORDS;
Query OK, 0 rows affected (0.02 sec)
```

8. Insert 3 records into the Authors table.

Answer:

```
mysql> INSERT INTO AUTHORS VALUES
    -> (101,"ROBER KIYOSAKI","USA"),
    -> (102,"NAPOLEON HILL","FRANCE"),
    -> (103,"RAJESH KADU","INDIA");
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

9. Insert 5 books into the Books table.

Answer:

```
mysql> INSERT INTO BOOKS VALUES
    -> (1,"DATABASE SYSTEMS",103,"2025",1000,"EDUCATION"),
    -> (2,"RICH DAD POOR DAD",101,"2008",5000,"FINANCE"),
    -> (3,"THINK AND GROW RICH",102,"2000",500,"SELF HELP"),
    -> (4,"BLOCKCHAIN TECHNOLOGY",103,"2026",100,"EDUCATION"),
    -> (5,"CYBER SECURITY",103,"2026",50,"EDUCATION");
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

10. Insert 3 members into the Members table.

Answer:

```
mysql> INSERT INTO MEMBERS VALUES
    -> (11,"SAIRAJ","sairaj123@gmail.com","9889450989"),
    -> (12,"AAYUSH","chalke98@gmail.com","8994212909"),
    -> (13,"SNEHA","sbhole@gmail.com","8167706460");
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

11. Insert 4 borrow records into the Borrow_Records table.

Answer:

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```
mysql> INSERT INTO BORROW_RECORDS VALUES (401,11,2,"2026-02-05","2026-02-10"),
    -> (402,12,5,"2025-12-19","2026-01-10"),
    -> (403,13,1,"2026-01-20","2026-02-10"),
    -> (404,11,3,"2024-06-21","2024-08-2");
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

12. Write an SQL query to select all books where published_year is after 2015.

Answer:

```
mysql> UPDATE BOOKS SET PUBLISHED_YEAR = "2015" WHERE BOOK_ID = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM BOOKS WHERE PUBLISHED_YEAR = 2015;
+---------+------------------+-----------+----------------+------------------+-----------+
| BOOK_ID | TITLE            | AUTHOR_ID | PUBLISHED_YEAR | AVAILABLE_COPIES | GENRE     |
+---------+------------------+-----------+----------------+------------------+-----------+
|       1 | DATABASE SYSTEMS |       103 |           2015 |             1000 | EDUCATION |
+---------+------------------+-----------+----------------+------------------+-----------+
1 row in set (0.00 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

Answer:

```
mysql> CREATE TABLE SCHOOL (HEAD VARCHAR(20),STANDARD INT PRIMARY KEY);
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE STUDENT(ID INT PRIMARY KEY,NAME VARCHAR(20),STANDARD INT,CONSTRAINT FK_STAN
DARD FOREIGN KEY (STANDARD) REFERENCES SCHOOL(STANDARD));
Query OK, 0 rows affected (0.05 sec)
```

14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

Answer:

```
mysql> SELECT M.*,B.TITLE FROM MEMBERS M JOIN BORROW_RECORDS BR ON M.MEMBER_ID = BR.MEMBER_ID JOI
N BOOKS B ON B.BOOK_ID = BR.BOOK_ID WHERE B.TITLE = "DATABASE SYSTEMS";
+-----------+-------+-----------------+------------+------------------+
| MEMBER_ID | NAME  | EMAIL           | PHONE      | TITLE            |
+-----------+-------+-----------------+------------+------------------+
|        13 | SNEHA | sbhole@gmail.com | 8167706460 | DATABASE SYSTEMS |
+-----------+-------+-----------------+------------+------------------+
1 row in set (0.00 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

Answer:

```
mysql> UPDATE BOOKS B JOIN BORROW_RECORDS BR ON B.BOOK_ID = BR.BOOK_ID SET AVAILABLE_COPIES = (A
VAILABLE_COPIES-1) WHERE B.BOOK_ID = BR.BOOK_ID;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

16. Delete a record from Members where member_id = 3.

Answer:

```
mysql> ALTER TABLE BORROW_RECORDS ADD CONSTRAINT FK_MEMBERID FOREIGN KEY (MEMBER_ID) REFEREN
CES MEMBERS(MEMBER_ID) ON DELETE CASCADE;
Query OK, 4 rows affected (0.13 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> DELETE FROM MEMBERS WHERE MEMBER_ID = 13;
Query OK, 1 row affected (0.02 sec)
```

17. Update a Book name record from Book table with id = 1.

Answer:

```
mysql> UPDATE BOOKS SET TITLE = "HADOOP SYSTEMS" WHERE BOOK_ID = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

18. Write an SQL query to list all books along with their authors' names.

Answer:

```
mysql> SELECT B.TITLE,A.NAME FROM BOOKS B JOIN AUTHORS A ON B.AUTHOR_ID = A.AUTHOR_ID;
+----------------------+----------------+
| TITLE                | NAME           |
+----------------------+----------------+
| RICH DAD POOR DAD    | ROBER KIYOSAKI |
| THINK AND GROW RICH  | NAPOLEON HILL  |
| HADOOP SYSTEMS       | RAJESH KADU    |
| BLOCKCHAIN TECHNOLOGY | RAJESH KADU    |
| CYBER SECURITY       | RAJESH KADU    |
+----------------------+----------------+
5 rows in set (0.00 sec)
```

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

Answer:

```
mysql> DELETE FROM BORROW_RECORDS WHERE BOOK_ID = 3;
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM BOOKS WHERE PUBLISHED_YEAR = 2000;
Query OK, 1 row affected (0.00 sec)
```

20. Write an SQL query to **find all books that are never borrowed (i.e., no records in `Borrow_Records`).**

Answer:

```
mysql> SELECT B.BOOK_ID,B.TITLE FROM BOOKS B LEFT JOIN BORROW_RECORDS BR ON B.BOOK_ID = BR.BOOK
_ID WHERE BR.BOOK_ID IS NULL;
+---------+----------------------+
| BOOK_ID | TITLE                |
+---------+----------------------+
|       1 | HADOOP SYSTEMS       |
|       4 | BLOCKCHAIN TECHNOLOGY |
+---------+----------------------+
2 rows in set (0.00 sec)
```